

Yiran Hu

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EDUCATION

Purdue University, West Lafayette, IN Aug 2023 – May 2028 (Expected)
Ph.D. in Computer Science · Advisor: [Prof. Bharat Bhargava](#)
Tianjin University, Tianjin, China Sep 2019 – Jun 2023
B.S. in Computer Science and Technology · GPA: 89.45/100 (3.73/4.0) · Advisor: [Prof. Junjie Chen](#)

PUBLICATIONS

- [P1] **Analyzing and Mitigating Cost-Inefficient Behaviors in Coding Agents** Under Review
Yiran Hu, Nan Jiang, Shanchao Liang, Anik Dey, Yi Wu, Lin Tan. [\[PDF\]](#)
- [C1] **TENET: One Step Toward Test-Driven Development for Repository-Level Code Generation** ISSRE 2026
Yiran Hu, Nan Jiang, Shanchao Liang, Yi Wu, Lin Tan. [\[PDF\]](#)
- [C2] **Can Language Models Replace Programmers for Coding? REPOCOD Says ‘Not Yet’** ACL 2025
Shanchao Liang, Nan Jiang, Yiran Hu, Lin Tan. [\[PDF\]](#)
- [C3] **SELP: Generating Safe and Efficient Task Plans for Robot Agents with LLMs** ICRA 2025
Yi Wu, Zikang Xiong, Yiran Hu, Shreyash S. Iyengar, Nan Jiang, Aniket Bera, Lin Tan, Suresh Jagannathan. [\[PDF\]](#)
Best Paper Award Finalist
- [C4] **LocRDF: An Ontology-Aware Key-Value Store for Massive RDF Data** WISA 2022
Yunxiao Zhang, Jinghan Li, Xueyang Liu, Rong Cheng, Yiran Hu, Xin Wang. [\[PDF\]](#)

RESEARCH EXPERIENCE

Purdue University – Graduate Research Assistant Aug 2023 – Present

Analyzing and Optimizing Cost Efficiency of Coding Agents [P1, first author]

- Built an agent efficiency analysis pipeline and identified three cost-inefficient behaviors (subsumed retrieval, similar script generation, test re-execution) consuming up to 22.75% of task cost in Claude Code and Mini-SWE-Agent.
- Optimized agent cost efficiency using structure-aware retrieval (CodeGraph) and agent-synthesized skills, over 10k+ agent runs in parallel containers on SWE-bench Verified and Pro, finding inconsistent savings.
- Designed developer-written agent skills to improve agent behavioral efficiency, cutting task cost by up to 41.73% while maintaining Pass@1.

TENET: Test-Driven Coding Agent for Repository-Level Code Generation [C1, first author]

- Designed a test-driven agentic framework for repository-level code generation, whose test harness selects a concise test subset by caller diversity and invocation proximity, outperforming the strongest baseline by 9.49 pp in Pass@1.
- Built a retrieval (Tree-sitter AST) and debugging (pdb) toolset for navigating large codebases and diagnosing failures; removing it drops Pass@1 by 14.89 pp.
- Designed a reflection-based refinement workflow that analyzes test failures to iteratively revise code; removing it drops Pass@1 by 9.24 pp.

REPOCOD: Repository-Level Code Generation Benchmark [C2]

- Curated 980 code-generation tasks from 11 popular Python repositories with developer-written tests to benchmark LLMs on realistic, repository-level coding tasks.
- Built repository-specific Docker environments for reproducible, execution-based evaluation that double as verifiable-reward environments for agent training.
- Evaluated commercial LLMs with RAG (including BM25) to compare different retrieval approaches, showing none of the 10 LLMs exceeded 30% Pass@1.

SELP: Safe and Efficient LLM Planners for Robot Agents [C3]

- Implemented an equivalence-voting mechanism to make LLM-generated Linear Temporal Logic (LTL) specifications more reliable, raising translation accuracy from 88.4% to 98.0%.
- Synthesized drone-navigation and tabletop-manipulation datasets via grammar-based LTL generation and LLM paraphrasing to train LLM planners on complex, safety-constrained tasks.

- Contributed to supervised fine-tuning (SFT) of Llama-family translators and planners on the synthesized data; SELP improves safety rate by up to 20.4% over prior LLM planners.

Software Analysis and Intelligence Lab, Tianjin University – Undergraduate Researcher Nov 2021 – May 2023

- Adapted the LEMON, deep-learning library testing tool from TensorFlow to MindSpore, uncovering 8 bugs.
- Applied differential testing to the LLVM and TVM compilers to detect compiler bugs.
- Optimized ResNet-50 with TVM graph- and operator-level tuning, reducing inference latency by 14.4 ms.

INDUSTRY EXPERIENCE

Zhongke Wenge Science and Technology Co., Ltd., Beijing, China

May 2022 – Aug 2022

Algorithm Engineer Intern, Research & Development Center

- Diagnosed PaddleOCR failure modes on COCO-Text and SVT (background interference, dense text, curved layouts) to locate accuracy bottlenecks.
- Integrated SAST detection with SVTR recognition trained on ICDAR2015 to handle distorted text, improving detection accuracy by 27.8%.
- Added a recognition module for detectors previously incompatible with recognition, improving accuracy by 15.4%.

COURSE PROJECTS

Test-Time Compute Scaling for LLM Math Reasoning

Spring 2026

- Compared Self-Consistency and PRM-weighted Best-of-N on MATH-500 (Llama-3.2-1B/3B, Llama-3.1-8B PRM, $n=1-64$) to study when a reward model beats majority voting.
- Found that correct answers exist in 73%–81% of candidate pools but majority voting mechanism in Self-Consistency often misses them; PRM-weighted selection beats it by 5.4 pp on the 1B model.

TEACHING

Teaching Assistant, Purdue University

Aug 2023 – Present

CS 176 Data Engineering in Python · CS 251/253 Data Structures & Algorithms · CS 571 Artificial Intelligence

HONORS & AWARDS

Best Paper Award Finalist , IEEE International Conference on Robotics and Automation (ICRA)	2025
Outstanding Graduate, Tianjin University	2023
First Prize, Challenge Cup National College Student Sci-Tech Competition (Tianjin University round)	2022
First Class Scholarship, Tianjin University	2021

SKILLS

Languages: Python, C++

LLM Agents: Claude Code, Agent Skills, Google ADK, Mini-SWE-Agent, OpenHands, MCP

Agent System Design: Agent Orchestration, Refinement Loop, Toolset Design, Prompt Engineering, RAG

LLM & ML: Anthropic/OpenAI APIs & SDKs, Hugging Face, PyTorch, TensorFlow, Data Synthesis, SFT

Evaluation: Code Generation Benchmarks, SWE-bench, Agent Trajectory Analysis, Cost Efficiency Analysis

Infrastructure: Docker, Linux, Git, Pytest, Tree-sitter